

RESEARCH ARTICLE

Effects of artichoke leaf extract supplementation on metabolic parameters in women with metabolic syndrome: Influence of TCF7L2-rs7903146 and FTO-rs9939609 polymorphisms

Khatereh Rezazadeh¹ | Mohammad Rahmati-Yamchi² | Leila Mohammadnejad³ |
Mehrangiz Ebrahimi-Mameghani⁴  | Abbas Delazar⁵

¹Talented Students Center, Student Research Committee, School of Nutrition and Food Sciences, Tabriz University of Medical Sciences, Tabriz, Iran

²Drug Applied Research Center, Tabriz University of Medical Sciences, Tabriz, Iran

³Immunology Research Center, Tabriz University of Medical Sciences, Tabriz, Iran

⁴Nutrition Research Center, School of Nutrition and Food Sciences, Tabriz University of Medical Sciences, Tabriz, Iran

⁵Department of Pharmacognosy, School of Pharmacy, Tabriz University of Medical Sciences, Tabriz, Iran

Correspondence

Mehrangiz Ebrahimi-Mameghani, Nutrition Research Center, School of Nutrition and Food Sciences, Tabriz University of Medical Sciences, Attar Neyshaboori Av., Golghasht St., Tabriz, Iran.
Email: ebrahimimamagani@tbzmed.ac.ir

The metabolic syndrome (MetS) is a multicomponent condition with a complex etiology involving genetic and environmental factors. Artichoke leaf extract (ALE) has shown favorable effects on lipid and glucose metabolism. The present study aimed to investigate the effects of ALE supplementation on metabolic parameters in women with MetS, using a nutrigenetics approach. In this double-blind randomized clinical trial, 50 women (aged 20–50 years) with MetS were randomly allocated into the two groups: “ALE group” (received 1,800 mg hydroalcoholic extract of artichoke as four tablets per day) and “placebo group” (received placebo consisted of corn starch, lactose, and avicel as four tablets per day) for 12 weeks. The biochemical and anthropometric parameters were determined before and after the intervention. The FTO-rs9939609 and the TCF7L2-rs7903146 polymorphisms were genotyped by polymerase chain reaction-restriction fragment length polymorphism. In carriers of A allele of the FTO-rs9939609, ALE supplementation resulted in a statistically significant decrease in serum triglyceride level compared with placebo (−19.11% vs. 10.83%; $p < .05$), with no other significant differences between the two groups. The TCF7L2-rs7903146 polymorphism showed no interaction with response to ALE ($p > .05$). These findings suggest that ALE supplementation may improve serum triglyceride level in A allele genotype of FTO-rs9939609 polymorphism in women with MetS.

KEYWORDS

artichoke leaf extract, FTO, metabolic syndrome, nutrigenetics, TCF7L2

1 | INTRODUCTION

Globe artichoke (*Cynara cardunculus* var. *scolymus* L.) belongs to Asteraceae family and is considered as a healthy food, which is widely consumed around the world. Extracts from artichoke leaves have been used in traditional medicine as a remedy to treat hepatobiliary and dyspeptic diseases (Lagow, 2004). In experimental studies and some clinical trials, artichoke leaf extract (ALE) has indicated several therapeutic properties such as antioxidant activity (Gebhardt, 1997; Rezazadeh,

Aliashrafi, Asghari-Jafarabadi, & Ebrahimi-Mameghani, 2017), hepatoprotective effects (Gebhardt, 1997), and glucose and lipid-lowering activity (Fantini et al., 2011; Qiang, Lee, Ye, Wu, & Hendrich, 2012; Rondanelli et al., 2014; Shimoda et al., 2003; Wider, Pittler, Thompson-Coon, & Ernst, 2013). ALE has shown good tolerability with rare adverse effects in clinical trials without any toxic effects in animal models (Rondanelli et al., 2014; Wider et al., 2013; Zan et al., 2013).

The main active components of ALE are flavonoids (e.g., luteolin and luteolin-7-O-glucoside), caffeic acid, mono- and dicaffeoylquinic acids (e.g., cynarin and chlorogenic acid) and bitters (sesquiterpenes lactones) (LaGow, 2004). The hypoglycemic effects of ALE seem to be mostly related to dicaffeoylquinic acid derivatives and chlorogenic acid (Arion et al., 1997; Matsui, Ogunwande, Abesundara, & Matsumoto, 2006). Previous studies have shown that the cholesterol lowering properties of ALE can be mediated by an increase in biliary

Abbreviations: MetS, metabolic syndrome; WC, waist circumference; HC, hip circumference; HDL-C, high-density lipoprotein cholesterol; TG, triglyceride; FBS, fasting blood sugar; LDL-C, low-density lipoprotein cholesterol; TC, total cholesterol; SBP, systolic blood pressure; DBP, diastolic blood pressure; FTO, fat mass and obesity associated; TCF7L2, transcription factor 7-like 2; ALE, artichoke leaf extract

secretion and a reduction of cholesterol biosynthesis through the inhibition of hepatic HMG-CoA reductase; this inhibitory effect was mainly attributable to luteolin (Kraft, 1997).

Metabolic syndrome (MetS) is a constellation of cardiometabolic abnormalities associated with increased risk of cardiovascular disease and Type 2 diabetes. Although various definitions have been proposed by several major organizations including the World Health Organization (WHO), the National Cholesterol Education Program Adult Treatment Panel III (NCEP ATPIII) and the International Diabetes Federation (IDF) for the diagnosis of the MetS, the recent guidelines released by the joint interim statement 2009 defined hypertriglyceridemia, hypertension, dysglycemia, abdominal obesity, and decreased high-density lipoprotein cholesterol (HDL-C) levels as the main criteria (Alberti et al., 2009). Abdominal obesity is the central feature of the MetS. Recent evidence showed that obesity-induced low-grade inflammation is related to insulin resistance (IR), a linking factor in the pathophysiology of the MetS (De Luca & Olefsky, 2008). In addition, mitochondrial dysfunction, caused by overproduction of reactive oxygen species in patients with MetS, plays a key role in developing IR (Kim, Wei, & Sowers, 2008). MetS results from an interplay of environmental factors such as increasing obesity and sedentary lifestyle and genetic factors, where heritability of MetS might differ between 10% and 30% (Alberti et al., 2009). It has been suggested that lifestyle modifications such as exercise and calorie restriction to reduce obesity could improve IR and consequently MetS through enhancement of mitochondrial function (Kim et al., 2008).

A recent meta-analysis reported that polymorphisms in transcription factor 7-like 2 (TCF7L2) and fat mass and obesity-associated (FTO) genes are major candidate genes for predisposing to MetS (Povel, Boer, Reiling, & Feskens, 2011). The single nucleotide polymorphism (SNP) rs7903146 of the TCF7L2 gene, located between Exons 4 and 5 on chromosome 10, plays a central role in pancreatic beta and alpha cell metabolism. It is demonstrated that individuals carrying T allele of rs7903146-TCF7L2 are at increased risk for both Type 2 diabetes and MetS. Moreover, the effect of the TCF7L2 variant on homeostasis of blood glucose might be related to the regulation of proglucagon gene expression (Povel et al., 2011; Sjögren et al., 2008). The SNP rs9939609 of the FTO gene, located within the first FTO intron, has been shown to be associated with an increased risk of obesity, Type 2 diabetes, and MetS (Freathy et al., 2008; Povel et al., 2011; Xiang et al., 2016). There is also evidence to indicate its relationship with four components of MetS including waist circumference (WC), fasting blood sugar (FBS), triglyceride (TG), and HDL-C dependently of body mass index (BMI). The exact mechanism of action of FTO variant is not clear but could be mediated by body weight regulation (Freathy et al., 2008).

Although the potential positive effects of ALE consumption on lipid and glucose metabolism have been already identified in animal and in vitro studies, results of limited clinical trials are discrepant. Moreover, genetic differences in variability in response to ALE supplementation have not been ever studied. Therefore, we aimed to examine the effects of ALE supplementation on anthropometric and metabolic parameters in women with MetS and to study the effects of the rs7903146 polymorphism in the TCF7L2 gene and the rs9939609 polymorphism in the FTO gene on the metabolic parameters response to ALE supplementation.

2 | MATERIALS AND METHODS

2.1 | Participants

Fifty-six women with MetS were recruited through printed advertisements and referrals by clinicians. The women with MetS were identified based on the international criteria, as having three traits out of five (Alberti et al., 2009). The following criteria were used to MetS diagnosis: HDL-C < 50 mg/dl, FBS $\geq$ 100 mg/dl, systolic blood pressure (SBP) $\geq$ 130 and/or diastolic blood pressure (DBP) $\geq$ 85 mg/dl, TG $\geq$ 150 mg/dl, and WC $\geq$ 95 cm. The reported WC cut points by the Iranian National Committee of Obesity were used to identify abdominal obesity (Azizi et al., 2010).

Women with self-reported sprue and Crohn's disease, liver dysfunction, biliary tract obstruction and gallstones, inflammatory disease, hypothyroidism or hyperthyroidism, cancer, renal failure, diabetes mellitus, cardiovascular diseases, and recent surgery were excluded. Moreover, taking antioxidant supplements (selenium, lycopene, carotenoids, vitamin E, vitamin C, and fish oil), consuming corticosteroids, antihypertensive, and fat-lowering medications over the past 3 months, following a weight-loss diet, being hypersensitive to artichoke, as well as pregnancy, lactation, and menopause were also considered as exclusion criteria.

The study was approved by the Ethics Committee of Tabriz University of Medical Sciences (reference number: 93120), and written informed consent was obtained from each subject. The trial was registered on the Iranian Registry of Clinical Trials (<http://www.irct.ir/>, IRCT201409033320N9).

2.2 | Study design and randomization

This study was a parallel, randomized, double-blind, placebo-controlled clinical trial conducted in Khoy, Iran, from November 2014 to May 2015. The sample size was calculated based on the study by Rondanelli et al. (2014) and considering the changes in serum glucose level. In each group, at least 23 patients were needed to provide an 80% power and a 95% confidence level. The sample size was increased to 28 women with MetS aged 20 to 50 years per group by considering a 20% dropout. The patients were randomly allocated into the two groups: ALE and placebo. The allocation was performed using permuted block randomization procedure by Random Allocation Software, based on age matching. A statistician with no clinical involvement in the study conducted the allocation. Prescreening was applied over the phone for all volunteers, and then eligible patients were entered into the study after anthropometric and biochemical assays. The patients were assigned to receive four tablets of ALE or placebo per day (one before breakfast, one before dinner, and two before lunch) for 12 weeks. Each tablet contained 450 mg ALE or placebo, provided in similar size and color in opaque plastic bottles. The patients were asked to return the remaining tablets at each visit for evaluation of compliance. They were advised to keep their usual dietary intake and physical activity throughout the study and to inform the researchers about any possible adverse effects related to the supplements.

2.3 | Supplements preparation

Fresh globe artichoke (*Cynara scolymus* L.) leaves were obtained from the Iran's medicinal plants cultivation company. ALE and placebo tablets were prepared by Dineh Iran Co (Qazvin-Iran). The alcoholic extraction (EtOH 70%) was applied to prepare the hydroalcoholic extract of artichoke leaf. The extract contained at least 4–5% chlorogenic acid, which was standardized using the spectrophotometer technique at 505 ± 2 nm, based on the protocol introduced by Iranian Herbal Pharmacopoeia (Ghasemi, 2002). The content of placebo tablet was inactive ingredient such as corn starch, lactose, and avicel.

2.4 | Anthropometric indices and blood pressure measurement

Anthropometric indices and blood pressure were measured at baseline and at the end of the study. The patients were weighed using a digital scale (Omron BF511, Kyoto, Japan) with a precision of 100 g with subjects wearing light clothing with no shoes. Height of the patients was measured without shoes using a wall-mounting measuring tape (Soehnle 5002.01, Hamburg, Germany) with a precision of 1 mm. Then, BMI was calculated ($\text{BMI} = \text{weight [kg]} / \text{height [m]}^2$). WC measurement was performed at the halfway between the lowest rib and the iliac crest with a precision of 1 mm. Hip circumference (HC) was measured at the maximal gluteal circumference. Waist-to-hip ratio was also determined by dividing WC by HC. Blood pressure (BP) was measured using an automated digital sphygmomanometer (Microlife A100–30, Berneck, Switzerland) after 5 min of rest in a seated position. The measurement was repeated with a 5-min interval; the mean of two measurements was taken as BP.

2.5 | Dietary intake and physical activity measurement

Dietary intake was assessed by three 24-hr food recalls (one weekend and two working days) before and after the supplementation period. Dietary energy and nutrients intakes were analyzed using Nutritionist IV Software modified for Iranian foods (First Databank; Hearst, San Bruno, CA, USA). The short form of the International Physical Activity Questionnaire was used for assessment of physical activity levels. Based on the scoring method of International Physical Activity Questionnaire, those participants who performed activities achieving a minimum total physical activity of at least 3,000 metabolic equivalents task (MET)-min/week, at least 600 MET-min/week, and lower than 600 MET-min/week was classified as “high,” “moderate,” and “low” active, respectively (IPAQ Research Committee, 2005; Craig et al., 2003).

2.6 | Biochemical measurements

Fasting blood samples were collected from all patients before and after the intervention. Centrifugation at a 3,000 rpm for 10 min was used to separate serum samples. Serum levels of lipid profile and glucose were determined on the day of blood sampling, and the remnants of sera were stored at -80°C until assay time. The concentrations of serum TG, TC, HDL-C, and FBS were measured using the standard enzymatic-colorimetric method on an automatic biochemical Hitachi 717

analyzer (Hitachi, Boehringer Mannheim, Japan) by Pars-Azmoon kits (Tehran, Iran). Friedewald formula was used to estimate the LDL-C level: $\text{LDL-C} = \text{TC} - (\text{HDL-C} + \text{TAG}/5)$; Friedewald, Levy, & Fredrickson, 1972). Serum insulin concentration was assayed using human ELISA kit (Monobind Inc., Lake Forest, CA, USA). Insulin resistance was determined by the homeostasis model assessment of insulin resistance (HOMA-IR; Haffner, Gonzalez, Miettinen, Kennedy, & Stern, 1996), as well as insulin sensitivity was assessed by the quantitative insulin sensitivity check index (Katz et al., 2000).

2.7 | Genotyping

Genomic DNA was extracted from whole-blood samples, which had been stored at -80°C in tubes containing anticoagulant ethylenediaminetetraacetic acid using the rapid DNA genomic extraction (RDGE) method. The single-nucleotide polymorphisms of the TCF7L2-rs7903146 and FTO-rs9939609 were genotyped using the polymerase chain reaction-restriction fragment length polymorphism method with mismatch primer modification as described earlier (Kowalska et al., 2012; Parra et al., 2007). The polymerase chain reaction product of the TCF7L2 gene was digested with a restriction enzyme, RsaI (Thermo Scientific, Vilnius, Lithuania), and the amplified product of the FTO gene

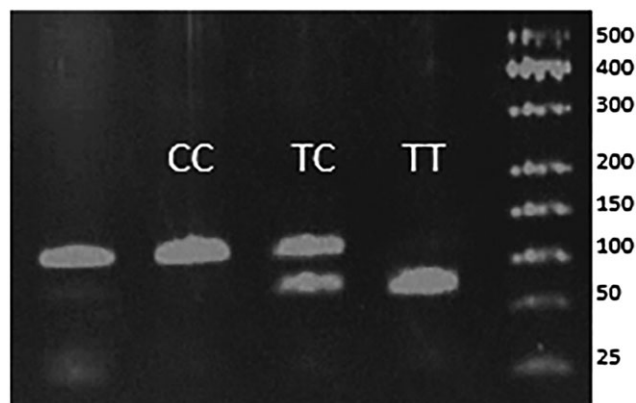


FIGURE 1 Polymerase chain reaction-based restriction fragment length polymorphism analysis of the TCF7L2-rs7903146: 25-bp DNA ladder.

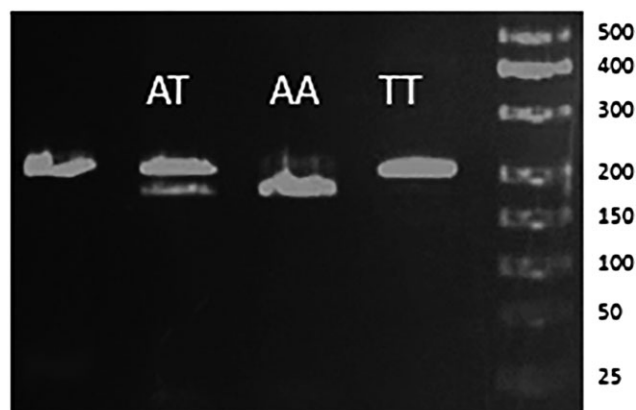


FIGURE 2 Polymerase chain reaction-based restriction fragment length polymorphism analysis of the FTO-rs9939609: 25-bp DNA ladder.

was digested with DdeI enzyme (Thermo Scientific, Vilnius, Lithuania). Digestion was performed according to the enzyme instruction and the digestion products were then separated on a 12% sodium dodecyl sulfate (SDS)-polyacrylamide gel electrophoresis and stained using ethidium bromide. The resulting fragments of the rs7903146 variant of TCF7L2 were 79-bp and 26-bp for the TT homozygotes; 105-bp, 79-bp, and 26-bp for the TC heterozygotes; and 105-bp for the CC homozygotes (Figure 1). Furthermore, genotyping of the rs9939609 variant of FTO resulted in 189-bp and 26-bp fragments for the AA homozygotes; 215-bp, 189-bp, and 26-bp bp fragments for the AT heterozygotes; and 215-bp undigested product for the TT homozygotes (Figure 2).

3 | STATISTICAL ANALYSIS

Data were analyzed on an intention-to-treat principle using SPSS software (ver. 16; SPSS Inc., IL, Chicago, USA). The data distribution was assessed by the Kolmogorov–Smirnov goodness of fit, and all variables had the normal distribution. At baseline, the independent sample *t* test and chi-square test were used to compare the means of continuous and categorical quantitative variables between the two groups, respectively. Moreover, the interaction between the TCF7L2-rs7903146 and the FTO-rs9939609 polymorphisms and intervention outcomes was analyzed using analysis of covariance test, adjusted for the baseline values. All quantitative data were reported as mean (standard deviation), mean difference (95% confidence interval), and number (percentage); $p < .05$ was as the threshold value of statistical significance.

3.1 | Results

3.2 | Characteristics of patients

Of the 230 volunteers, 172 were eligible and underwent anthropometric and biochemical assessment. Among them, 56 women with MetS were recruited in the trial. Over 12 weeks, three dropouts in ALE group and four dropouts in the placebo group occurred. Therefore, 49 patients completed the trial (Figure 3). There were no reports of adverse reactions caused by ALE or placebo consumption. The participants were premenopausal women with a mean (standard deviation) of 38.43 (6.43) years of age. Baseline anthropometric indices, dietary intake, and physical activity levels of patients in ALE and placebo groups were similar (Table 1). The allele frequency of TCF7L2-rs7903146 and FTO-rs9939609 polymorphisms had the same distribution between the two groups (Table 2).

3.3 | Genotype-intervention interaction

The modulation effects of TCF7L2-rs7903146 on anthropometric and biochemical profile in response to ALE supplementation are shown in Table 3. The baseline values of weight, BMI, WC, and HC in C allele carriers were different between the two groups ($p < .05$). The C allele carriers indicated a statistically significant decrease of WC, SBP, DBP, and FBS in ALE group and BPD, LDL-C, and FBS in the placebo group ($p < .05$). Moreover, the TT carriers revealed the significant reduction of WC, BMI, BPS, insulin, and HOMA-IR in ALE group and LDL-C in the placebo group ($p < .05$). However, there were no

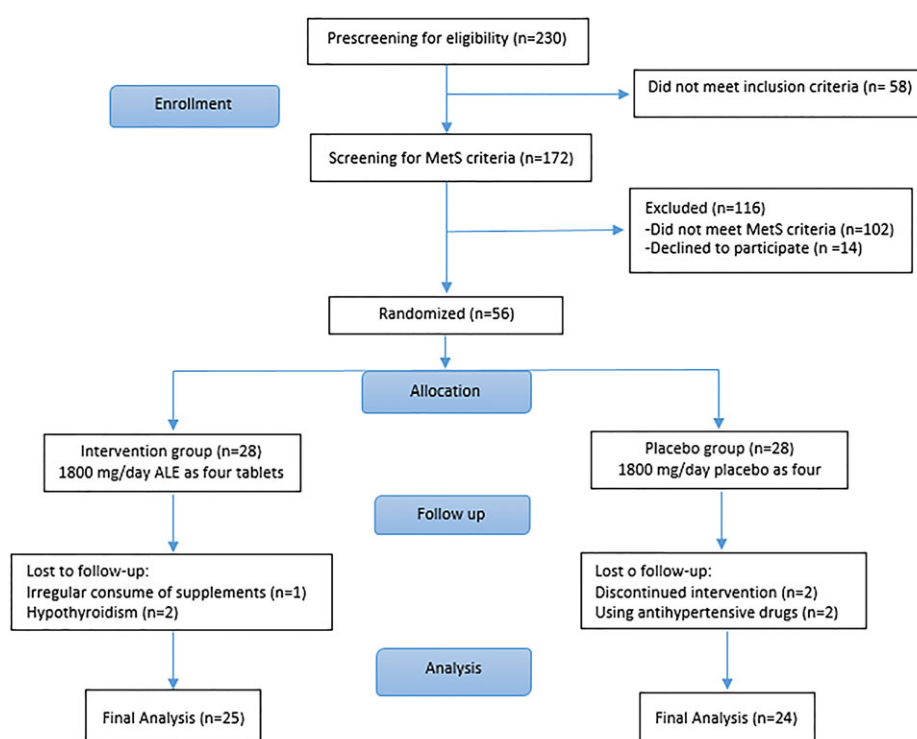


FIGURE 3 Flowchart of the study. ALE = artichoke leaf extract; MetS = metabolic syndrome. [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 1 Baseline characteristics of the study patients

Variables [#]	ALE group (n = 25)	Placebo group (n = 24)	p*
Age (year)	37.8 (7.4)	39.0 (5.3)	.519
Weight (kg)	91.1 (12.3)	81.4 (10.5)	.005
Height (cm)	158.3 (5.6)	157.1 (4.8)	.425
BMI (kg/m ²)	36.1 (4.7)	32.9 (3.6)	.012
Dietary energy (kcal/day)	1,883.6 (525.5)	1,755.6 (431.5)	.413
Dietary protein (g/day)	60.0 (18.2)	58.6 (13.1)	.775
Dietary carbohydrate (g/day)	317.5 (107.6)	280.7 (81.5)	.238
Dietary total fat (g/day)	45.2 (15.0)	46.6 (15.4)	.770
Dietary SFA (g/day)	9.7 (3.7)	12.2 (6.5)	.151
Dietary MUFA (g/day)	10.5 (4.3)	13.1 (5.5)	.113
Dietary PUFA (g/day)	13.0 (5.1)	12.3 (6.0)	.699
Physical activity			.708
Low	10 (20.4%)	7 (14.3%)	
Moderate	10 (20.4%)	12 (24.5%)	
High	5 (10.2%)	5 (10.2%)	

Note. BMI = body mass index; SFA = saturated fatty acids; MUFA = mono-unsaturated fatty acids; PUFA = polyunsaturated fatty acids; ALE = artichoke leaf extract.

*Independent samples t test for numeric variables and Pearson chi-square test for categorical variables.

[#]Mean (SD) for numeric variables and number (%) for categorical variables.

significant differences between the groups in TCF7L2-rs7903146 variants in response to ALE supplementation ($p > .05$).

The interaction between FTO-rs9939609 genotype and intervention is presented in Table 4. At the beginning of the study, the values of weight, BMI, WC, BPS, and BPD in A allele carriers of ALE group were higher than the placebo group ($p < .05$). At the end of the study, the within-group analysis in A allele carriers showed significant reductions in BPS, BPD, and TC in ALE group and BPD, LDL-C, and FBS in the placebo group. In addition, the patients with TT genotype experienced a statistically significant reduction in BPS in ALE group as well as BPS and LDL-C in the placebo group. The results of the analysis of covariance test, adjusting for baseline values, showed that ALE supplementation significantly decreased serum TG levels in patients with A allele genotype ($p < .05$), whereas the changes in other anthropometric and biochemical indices were similar between the two groups ($p > .05$).

In addition, after controlling for the confounding effects of BMI and physical activity level, the results did not change significantly (data not shown).

TABLE 2 Genotype allelic frequencies of the study patients regarding the single-nucleotide polymorphism of TCF7L2-rs7903146 and FTO-rs9939609

Study group	TCF7L2-rs7903146			p*	FTO-rs9939609			p*
	TT	TC	CC		AA	AT	TT	
ALE group (n = 25)	7 (14.3%)	14 (28.6%)	4 (8.2%)	.890	3 (6.1%)	16 (32.7%)	6 (12.2%)	.523
Placebo group (n = 24)	6 (12.2%)	15 (30.6%)	3 (6.1%)		1 (2.0%)	15 (30.6%)	8 (16.3%)	
Total (n = 49)	13 (26.5%)	29 (59.2%)	7 (14.3%)		4 (8.2%)	31 (63.3%)	14 (28.6%)	

Note. TCF7L2 = transcription factor 7-like 2; FTO = fat mass and obesity associated; ALE = artichoke leaf extract.

*Chi-square.

4 | DISCUSSION

The present study examined the modulation of metabolic response to ALE supplementation by FTO-rs9939609 and TCF7L2-rs7903146 polymorphisms in women with MetS. Over 12 weeks, a significant reduction in serum TG level was observed in carriers of A allele of the FTO-rs9939609 in ALE group as compared with controls. There was no genotype-intervention interaction for the TCF7L2-rs7903146 polymorphism.

The association between the A allele of the rs9939609 of FTO gene and increased risk of obesity and MetS has been demonstrated in cross sectional studies (Freathy et al., 2008; Povel et al., 2011), but there is limited knowledge about the influence of this polymorphism on intervention response. Several intervention studies have found an interaction between diet/lifestyle interventions and the rs9939609 polymorphism of FTO gene, although the results are inconsistent. Razquin et al. (2010) examined the effects of a Mediterranean-style diet on body weight changes in respect to FTO-rs9939609 in subjects at high cardiovascular risk. After 3 years of intervention, the subjects carrying the A allele had lower body weight gain compared with nonmutated subjects regardless of the type of nutritional intervention (Razquin et al., 2010). Luis et al. (2012) reported that FTO variant rs9939609 did not have an association with body weight response following low-fat or low-carbohydrate hypocaloric diets in obese patients over 3 months. However, A allele carriers had a better metabolic management secondary to weight loss with low-fat diet; the levels of TC, LDL-C, and c-reactive protein (CRP) significantly decreased in mutant type group compared with wild type group (Luis et al., 2012). Similarly, a recent meta-analysis showed that diet/lifestyle interventions might lead to more weight loss in subjects carrying A allele than noncarriers (Xiang et al., 2016). Müller et al. (2008) showed no association of FTO-rs9939609 with weight loss or metabolic parameters in obese children and adolescents (Müller et al., 2008). Lappalainen et al. (2009) did not find an association between FTO-rs9939609 and weight-loss response to a 4-year lifestyle intervention on patients with impaired glucose tolerance (Lappalainen et al., 2009).

Accordingly, some previous literature reported that subjects carrying risk allele of FTO-rs9939609 may respond better to intervention than noncarriers subjects (Luis et al., 2012; Razquin et al., 2010; Xiang et al., 2016), which are consistent with our results. Our findings revealed that A allele carriers experienced a significant improvement of TG level in ALE group compared with placebo group. However, few studies failed to show an association between FTO variant rs9939609 and response to intervention (Lappalainen et al., 2009;

TABLE 3 Effects of ALE on anthropometric and biochemical parameters, regarding TCF7L2-rs7903146 after 12 weeks

		TT (n = 13)		MD (95% CI), p*	C allele (n = 36)		MD (95% CI), p*
		ALE group (n = 7)	Placebo group (n = 6)		ALE group (n = 18)	Placebo group (n = 18)	
Weight (kg)	Before	86.6 (10.2)	78.4 (6.5)	1.1 (−0.1 to 2.3), .077	93.0 (12.8)	82.4 (11.5) [†]	−0.4 (−1.9 to 0.9), .514
	After	87.3 (10.3)	77.9 (6.6)		92.1 (11.9)	82.3 (11.9)	
BMI (kg/m ²)	Before	34.7 (5.3)	32.4 (1.9)	0.4 (−0.0 to 0.9), .063	36.6 (4.5)	33.0 (4.0) [†]	0.0 (−0.5 to 0.6), .915
	After	35.0 (5.3) [‡]	32.2 (2.0)		36.4 (4.2)	33.0 (4.1)	
WC (cm)	Before	106.0 (8.9)	98.5 (6.1)	0.8 (−0.8 to 2.6), .290	110.0 (6.0)	105.5 (5.3) [†]	−1.0 (−3.1 to 0.9), .288
	After	105.2 (7.9) [‡]	98.0 (4.7)		108.6 (6.0) [‡]	105.0 (7.2)	
HC (cm)	Before	116.8 (7.4)	114.5 (5.9)	1.3 (−0.2 to 2.9), .100	120.9 (8.8)	114.6 (8.3) [†]	−0.1 (−1.7 to 1.4), .846
	After	117.7 (7.7)	114.0 (6.0)		120.6 (8.9)	114.6 (8.5)	
Waist to hip ratio	Before	0.9 (0.0)	0.8 (0.0)	0.0 (−0.0 to 0.0), .728	0.9 (0.0)	0.8 (0.2)	0.0 (−0.0 to 0.0), .727
	After	0.8 (0.0)	0.8 (0.0)		0.9 (0.0)	0.9 (0.0)	
SBP (mmHg)	Before	128.5 (14.8)	123.5 (10.6)	−4.6 (−13.0 to 3.6), .241	132.1 (19.7)	121.9 (12.1)	−3.6 (−8.9 to 1.6), .173
	After	120.2 (16.2) [‡]	120.1 (10.0)		122.7 (17.2) [‡]	118.5 (11.2)	
DBP (mmHg)	Before	85.2 (9.8)	78.8 (7.1)	0.9 (−9.1 to 11.1), .836	82.1 (8.8)	78.4 (7.0)	0.2 (−3.0 to 3.5), .876
	After	78.8 (10.7)	73.1 (8.2)		78.8 (8.3) [‡]	75.5 (7.7) [‡]	
FBS (mg/dl)	Before	93.3 (5.6)	95.5 (9.3)	3.5 (−4.1 to 11.2), .326	105.0 (8.4)	101.2 (9.2)	−0.6 (−5.6 to 4.3), .792
	After	95.8 (7.5)	93.5 (6.3)		98.6 (8.7) [‡]	96.7 (8.9) [‡]	
Insulin (μU/ml)	Before	15.5 (6.8)	9.8 (4.2)	−6.8 (−15.7 to 1.9), .112	13.5 (6.6)	12.5 (3.7)	1.5 (−2.8 to 5.9), .480
	After	11.1 (4.0) [‡]	16.0 (8.2)		15.8 (5.6)	13.9 (7.3)	
HOMA-IR	Before	3.2 (1.4)	2.3 (1.1)	−1.4 (−3.3 to 0.5), .141	3.4 (1.6)	3.2 (1.1)	0.3 (−0.7 to 1.4), .526
	After	2.5 (1.0) [‡]	3.6 (1.6)		3.7 (1.5)	3.3 (1.6)	
QUICKI	Before	0.3 (0.0)	0.3 (0.0)	0.0 (−0.0 to 0.0), .260	0.3 (0.0)	0.3 (0.0)	0.0 (−0.0 to 0.0), .397
	After	0.3 (0.0)	0.3 (0.0)		0.3 (0.0)	0.3 (0.0)	
TG (mg/dl)	Before	211.2 (113.8)	218.6 (144.7)	1.6 (−54.2 to 57.6), .948	172.2 (82.7)	157.2 (47.1)	−30.0 (−71.8 to 11.7), .153
	After	170.1 (84.1)	174.3 (136.6)		147.2 (55.2)	171.3 (75.0)	
TC (mg/dl)	Before	215.4 (48.9)	229.7 (27.1)	10.8 (−27.4 to 49.1), .531	193.3 (30.6)	201.3 (24.3)	6.5 (−18.3 to 31.3), .595
	After	209.2 (41.2)	209.0 (37.6)		182.0 (31.7)	180.5 (39.9)	
LDL-C (mg/dl)	Before	119.2 (39.2)	122.5 (10.2)	16.9 (−7.3 to 41.2), .146	111.0 (22.96)	117.3 (22.9)	2.9 (−7.1 to 13.0), .551
	After	117.2 (32.9)	102.9 (21.6) [‡]		103.9 (23.45)	106.2 (22.4) [‡]	
HDL-C (mg/dl)	Before	47.4 (6.9)	43.3 (3.1)	3.9 (−6.7 to 14.6), .427	40.2 (8.7)	44.3 (8.4)	−0.3 (−5.8 to 5.1), .900
	After	49.9 (9.7)	43.9 (5.5)		42.8 (7.2)	45.3 (9.4)	

Note. Values are mean (SD). BMI = body mass index; WC = waist circumference; HC = hip circumference; FBS = fasting blood sugar; TG = triglyceride; HDL-C = high-density lipoprotein cholesterol; LDL-C = low-density lipoprotein cholesterol; TC = total cholesterol; SBP = systolic blood pressure; DBP = diastolic blood pressure; HOMA-IR = homeostasis model assessment for insulin resistance; QUICKI = quantitative insulin sensitivity check index; ALE = artichoke leaf extract; TCF7L2 = transcription factor 7-like 2; MD=mean difference; CI = confidence interval.

*Analysis of covariance (adjusted for baseline values).

[†]p < .05 for between group differences (independent samples t test), before of the intervention.

[‡]p < .05 for within group changes (paired t test).

Müller et al., 2008). These discrepancies in results might be due to the differences in the characteristics of the participants, type, and duration of interventions.

FTO is expressed in multiple tissues, especially in the hypothalamus, adrenal glands, and pituitary, but its biological function is not fully understood (Gerken et al., 2007). Previous studies demonstrated that FTO variants could play an important role in the regulation of energy balance and body weight (Xiang et al., 2016). However, the effects of the common FTO variant rs9939609 on metabolic parameters might be explained by the possible relationship of FTO with many metabolic pathways, which are largely unknown. Furthermore, as reported in our previous study (Rezazadeh et al., 2017), the evidence showed that ALE supplementation could decrease TG level (Qiang et al., 2012; Shimoda et al., 2003). Antihypertriglyceridemic effect of ALE might be mediated by active components of ALE (sesquiterpenes and sesquiterpene glycosides) and inhibition of gastric emptying (Shimoda et al., 2003).

Although the rs7903146 variant of TCF7L2 has shown strong association with Type 2 diabetes and hyperglycemia (Povel et al., 2011), there are limited trials that have studied the interaction between this polymorphism and response to intervention. In an earlier study by Corella et al. (2013), the Mediterranean diet modulated increased fasting lipids and glucose in TT genotype compared with C allele carriers after 4.8 years (Corella et al., 2013). Zimdahl et al. (2014) showed that 24-week treatment with linagliptin in patients with Type 2 diabetes resulted in less improvement of hemoglobin A1c in TT compared with CC genotype (Zimdahl et al., 2014).

TCF7L2 encodes a transcription factor involved in the regulation of blood glucose homeostasis. The probable mechanisms through which rs7903146-TCF7L2 increase the risk of diabetes and could affect the response to intervention are impaired insulin secretion or incretin effect (Lyssenko et al., 2007). However, our findings did not show an association between the rs7903146 variant of TCF7L2 and response to ALE supplementation. Our results are in agreement with

TABLE 4 Effects of ALE on anthropometric and biochemical parameters, regarding FTO-rs9939609 after 12 weeks

		A allele (n = 35)		MD (95% CI), p*	TT (n = 14)		MD (95% CI), p*
		ALE group (n = 19)	Placebo group (n = 16)		ALE group (n = 6)	Placebo group (n = 8)	
Weight (kg)	Before	91.0 (12.5)	81.2 (12.4) [†]	0.2 (−1.1 to 1.6), .722	91.6 (12.8)	81.7 (5.6)	−0.3 (−2.5 to 1.7), .695
	After	90.6 (11.6)	81.0 (12.7)		90.9 (11.9)	81.8 (6.3)	
BMI (kg/m ²)	Before	36.2 (4.6)	32.7 (4.2)	0.2 (−0.3 to 0.8), .354	35.6 (5.6)	33.3 (2.1)	−0.1 (−0.9 to 0.5), .608
	After	36.2 (4.3)	32.6 (4.3)		35.3 (5.3)	33.3 (2.2)	
WC (cm)	Before	109.2 (7.3)	104.1 (5.7) [†]	0.0 (−1.1 to 1.8), .933	107.6 (6.3)	102.8 (7.9)	−2.1 (−5.7 to 1.3), .200
	After	108.2 (8.2)	103.0 (7.0)		105.6 (4.3)	103.5 (8.5)	
HC (cm)	Before	119.5 (8.3)	114.4 (8.4)	0.7 (−0.7 to 2.1), .316	120.8 (9.8)	115.1 (6.3)	−0.9 (−3.8 to 1.9), .502
	After	119.7 (8.3)	114.0 (8.7)		120.0 (10.1)	115.3 (6.1)	
Waist to hip ratio	Before	0.9 (0.0)	0.9 (0.0)	0.0 (−0.0 to 0.0), .776	0.8 (0.0)	0.7 (0.2)	−0.0 (−0.0 to 0.0), .295
	After	0.9 (0.0)	0.9 (0.0)		0.8 (0.0)	0.8 (0.0)	
SBP (mmHg)	Before	133.3 (20.1)	121.0 (13.1) [†]	−3.0 (−8.5 to 2.4), .265	124.2 (8.3)	124.7 (7.7)	−6.3 (−14.4 to 1.8), .115
	After	124.5 (17.6) [‡]	117.9 (12.0)		114.1 (10.4) [‡]	120.8 (7.9) [‡]	
DBP (mmHg)	Before	84.2 (9.5)	76.6 (7.3) [†]	0.3 (−3.6 to 4.3), .844	79.3 (6.4)	82.2 (4.1)	−0.5 (−8.7 to 7.7), .893
	After	79.8 (8.8) [‡]	73.4 (8.3) [‡]		75.6 (8.7)	77.9 (5.6)	
FBS (mg/dl)	Before	101.7 (9.7)	100.2 (8.9)	0.7 (−3.9 to 5.4), .749	101.3 (8.0)	98.7 (10.8)	2.0 (−8.7 to 12.8), .675
	After	97.5 (9.0)	96.0 (6.8) [‡]		99.6 (2.0)	95.7 (11.1)	
Insulin (μU/ml)	Before	13.9 (6.4)	11.6 (2.9)	1.3 (−2.7 to 5.5), .500	14.5 (7.9)	12.2 (5.5)	−4.7 (−14.6 to 5.1), .314
	After	14.8 (6.0)	12.7 (5.3)		13.4 (4.4)	17.5 (9.8)	
HOMA-IR	Before	3.5 (1.5)	2.9 (0.8)	0.1 (−0.8 to 1.2), .718	2.4 (0.8)	3.0 (1.6)	−1.1 (−4.2 to 1.9), .402
	After	3.5 (1.5)	3.0 (1.3)		2.7 (0.9)	4.0 (2.0)	
QUICKI	Before	0.3 (0.0)	0.3 (0.0)	0.0 (−0.0 to 0.0), .963	0.3 (0.0)	0.3 (0.0)	0.0 (−0.0 to 0.0), .399
	After	0.3 (0.0)	0.3 (0.0)		0.3 (0.0)	0.3 (0.0)	
TG (mg/dl)	Before	189.6 (103.0)	166.8 (96.6)	−45.4 (−86.5 to −4.3), .031	162.6 (38.1)	184.1 (50.6)	18.1 (−52.8 to 89.2), .585
	After	153.4 (67.5)	184.8 (101.0)		154.5 (54.3)	146.6 (63.2)	
TC (mg/dl)	Before	200.0 (39.8)	204.1 (27.3)	−3.3 (−21.5 to 14.9), .711	201.3 (34.6)	216.8 (26.7)	39.0 (−20.5 to 98.6), .169
	After	184.1 (34.1) [‡]	190.8 (39.8)		208.3 (39.5)	176.4 (43.6)	
LDL-C (mg/dl)	Before	111.3 (29.5)	113.9 (21.5)	16.2 (−2.0 to 34.4), .076	118.4 (26.7)	125.7 (17.3)	2.3 (−8.7 to 13.4), .663
	After	103.5 (24.8)	103.2 (23.2) [‡]		118.0 (29.7)	108.6 (20.2) [‡]	
HDL-C (mg/dl)	Before	41.3 (9.8)	43.0 (5.8)	−0.9 (−6.6 to 4.7), .738	45.5 (4.1)	45.6 (9.3)	5.5 (−2.7 to 13.7), .168
	After	43.2 (8.3)	45.2 (9.5)		49.9 (7.7)	44.4 (6.8)	

Note. Values are mean (SD). BMI = body mass index; WC = waist circumference; HC = hip circumference; FBS = fasting blood sugar; TG = triglyceride; HDL-C = high-density lipoprotein cholesterol; LDL-C = low-density lipoprotein cholesterol; TC = total cholesterol; SBP = systolic blood pressure; DBP = diastolic blood pressure; HOMA-IR = homeostasis model assessment for insulin resistance; QUICKI = quantitative insulin sensitivity check index; ALE = artichoke leaf extract; FTO = fat mass and obesity associated; MD = mean difference; CI = confidence interval.

*Analysis of covariance (adjusted for baseline values).

[†]p < .05 for between group differences (independent samples t test), before of the intervention.

[‡]p < .05 for within group changes (paired t test).

those of Mattei, Qi, Hu, Sacks, and Qi (2012), who examined the interaction between two variants (rs7903146 and rs12255372) of TCF7L2 and 2-year weight-loss diets with different macronutrient proportions. Although, significant interactions were observed for TCF7L2-rs12255372 and fat intake for changes in the body adiposity, no significant interactions were reported for TCF7L2-rs7903146 (Mattei et al., 2012).

Previous clinical trials have shown the efficacy of ALE in serum glucose regulation (Rondanelli, Giacosa, Orsini, Opizzi, & Villani, 2011; Rondanelli et al., 2014). Some active ingredients of ALE have been proposed for its glucose lowering effects. The dicaffeoylquinic acid derivatives may modulate the activity of alpha-glucosidase, an enzyme responsible for the catabolism of dietary carbohydrates, and chlorogenic acid could inhibit glucose 6-phosphate translocase, which is a component of the glucose 6-phosphatase complex (Arion et al., 1997; Matsui et al., 2006). Flavonoids are other possible ingredients of ALE, which can exhibit antidiabetic effects. Although flavonoids

have been reported to protect pancreatic β-cells against cytokine-induced damage (Kim et al., 2007), hypoglycemic effects of flavonoids in ALE have not been demonstrated. In addition, flavonoids possess a variety of pharmacological effects such as antioxidant, anticancer, antiviral, and hepatoprotective activities (Nijveldt et al., 2001; Seelinger, Merfort, & Schempp, 2008). Furthermore, they play an antiinflammatory role through the antioxidant effects, inhibition of proinflammatory substances and suppression of the nuclear factor kappa B pathway (Chen, Teng, Jia, & Battino, 2017).

In our study, ALE supplementation failed to affect FBS, HOMA-IR, and quantitative insulin sensitivity check index (QUICKI), whereas, the interaction between the rs7903146 variant of TCF7L2 and glucose and insulin indices was found in those studies that the intervention has led to the improvement of these indices (Corella et al., 2013; Zimdahl et al., 2014). Therefore, the different response of fasting glucose, insulin resistance, and insulin sensitivity to intervention may have accounted for the discrepancy between our results and others.

Moreover, differences in study duration, sample size, and baseline clinical characteristics of the subjects may also play a role in inconsistent results. In addition, we studied only one SNP of TCF7L2, whereas other SNPs of this gene such as rs12255372 also revealed interaction with intervention (Mattei et al., 2012).

Currently, there is considerable interest in the use of medicinal plant extracts as effective agents for the treatment of the MetS (Graf, Raskin, Cefalu, & Ribnick, 2010). In a study by Ziegenfuss, Hofheins, Mendel, Landis, and Anderson (2006), 22 subjects with the MetS consumed a water-soluble cinnamon extract for 12 weeks. The supplementation led to a significant decrease in FBS, SBP, and a significant increase in lean mass (Ziegenfuss et al., 2006). Sivaprakasapillai, Edirisinghe, Randolph, Steinberg, and Kappagoda (2009) examined the effects of grape seed extract in 27 patients with the MetS for 4 weeks. Although a reduction of systolic and diastolic blood pressures were observed in grape seed extract group as compared with placebo, the changes of lipids or blood glucose levels were not significant (Sivaprakasapillai et al., 2009). Gurrola-Díaz et al. (2010) conducted a factorial, randomized, follow-up study on the effects of *Hibiscus sabdariffa* extract powder (HSEP) in subjects with and without the MetS. After 4 weeks, glucose levels, cholesterol levels, and TG/HDL-C ratio significantly decreased, and HDL-C levels significantly increased in patients with MetS treated with HSEP. Moreover, TG levels reduced significantly in MetS patients treated with HSEP plus a recognized preventive treatment (diet) and subjects without MetS treated with HSEP (Gurrola-Díaz et al., 2010). Our results also indicated the TG-lowering effects of ALE in women with MetS. Therefore, medicinal plant extracts have shown favorable effects in improving some features of the MetS.

To the best of our knowledge, the present double-blind randomized clinical trial found for the first time that the common FTO variant rs9939609 may influence serum TG-level response to ALE supplementation in women with MetS. However, this study had some limitations: (a) only women were included; (b) duration of intervention was relatively short (only 12 weeks); and (3) standardization of ALE was performed using spectrophotometry, a technique with less precise than high-performance liquid chromatography, due to funding constraints.

5 | CONCLUSION

In women with MetS, ALE supplementation resulted in a significant improvement in serum TG levels in A allele carriers of rs9939609 polymorphism of FTO gene, a group at an increased risk of obesity and MetS. However, rs7903146 polymorphism of TCF7L2 gene did not show any significant interaction with response to ALE supplementation. These findings highlight the need for further long-term randomized clinical trials in a larger sample size to identify which group of patients with MetS may particularly benefit from ALE supplementation.

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CONFLICT OF INTEREST

The authors have declared that there is no conflict of interest.

ORCID

Mehrangiz Ebrahimi-Mameghani  <http://orcid.org/0000-0002-0311-1289>

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